Referee report, AER-2023-1106,
"Back to the 1980s or Not? The Drivers of Inflation and Real Risks in Treasury Bonds"

Summary: The paper studies the stock-bond correlation with the objective of learning what types of economic shocks (supply, demand) and what type of Taylor rule (weights, inertia) is driving the economy. The focus is on whether and when the bond market is perceived as risky, assessed based on its correlation with the stock market.

The motivating fact is in Figure 1, Panel A, which shows in red the stock-bond correlation for nominal bonds. This was positive in the 1980s and over most of the 1990s. It then turned negative in the 2000s. However, post-covid, the correlation has remained negative, despite seemingly large supply shocks during both the 1980s & 1990s as well as the post-covid period.

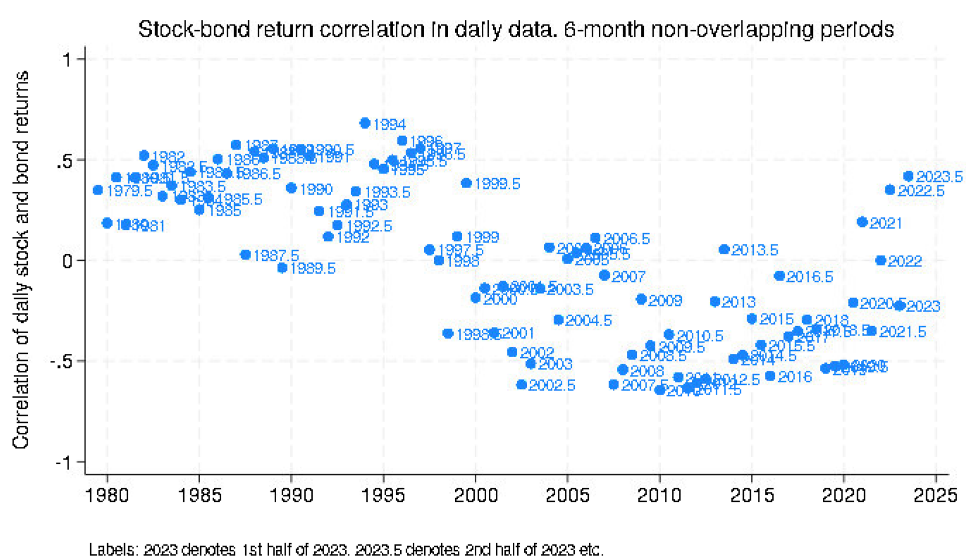
The central message is that monetary policy changes the effect of supply shocks on the stock-bond correlation. Specifically, monetary policy shifted from a "quick-acting and inflation focused monetary policy rule in the 1980s to an inertial and more output-focused monetary policy rule in the 2000s" and 2000s-style monetary policy continued into the post-covid period and changed the effect of supply shocks. As shown in the middle column in Figure 6, when 2000s-style monetary policy lets the real rate fall in response to inflationary supply shocks, then (a) supply shocks are good for stocks which benefits from the lower real yield, and (b) supply shocks are still bad for bonds (though less than for 1980s monetary policy, since the increase in nominal long bond yields is smaller). Because of (a), the stock-bond return correlation due to supply shocks switches from positive to negative, thus explaining the seemingly puzzling lack of a positive correlation post-covid.

The argument is made in the context of the model very similar to that in Campbell, Pflueger and Viceira (2020). The model that implies the standard three New Keynesian model equations: A forward and backward-looking Euler equation (eqn. 17) for the output gap x is derived from habit formation preferences and assumptions about the habit process and the output gap. A wage Phillips curve (eqn. 20) comes from wage-setting unions' first-order condition, and a standard inertial Taylor rule (eqn. 24) is assumed for monetary policy. Stock and bond pricing follows from the stochastic discount factor based on habit-formation preferences. A bond preference shock is used to generate demand shocks. Model parameters are set by a mix of calibration and Simulated Method of Moments. Counterfactual parameters are used to show how the model-implied stock-bond correlation changes with the nature of the stocks and the monetary policy rule.

Comments:

(1) Omission of recent data. Link to soft landings

The paper's motivating question is (first sentence): "What does data from nominal and real Treasury bond risks tell us about supply shocks and stagflation or, conversely, the Fed's ability to engineer a "soft landing"?" It concludes (last paragraph): "In particular, when the economy is driven by volatile supply shocks, nominal bond stock betas in the model emerge as a forward-looking indicator of "soft landings"". Are we to conclude that markets think the Fed will succeed in achieving a soft landing because the stock bond correlation post-covid has not been much above zero? What cutoff for the correlation would suggest success? And how do we then interpret the more recent evidence: The stock-bond correlation has been substantially positive after the end of the paper's sample which ends in June 2022. Here is a quick chart based on data from FRED on the Willshire 5000 for stocks and the 10-year Treasury for bonds:



The paper ends the sample in June 2022, corresponding to the dot labeled 2022 (first half of 2022) in my figure above. It thus omits July 2022 onward, which has had two half years with very positive stock-bond correlations and one with a negative correlation (that period is the first half of 2023 which includes the period of the March regional bank issues when bonds did well as yields fell, but stocks did poorly). The author states that "I end the sample in 30/06/2022, when the conduct of monetary policy clearly changed". What exactly changed about monetary policy? It got less inertial? The weight on inflation went up? Are we to conclude that the market thinks the monetary policy rule is now back to a 1980s rule which implies that a soft landing is *not* likely? This seems central to the paper's message which would then be that post-covid, at first the market thought the Fed could achieve a soft landing but it lost faith in this sometime in mid/late 2022, perhaps due to the Fed's aggressive tightening.

(2) What can one learn from the stock-bond return correlation about shocks and monetary policy?

From Figure 6, I learned the following:

- In the 1979-2001 calibration, all three shocks (supply shocks, demand shocks and monetary policy shocks) lead to a *positive* stock-bond return correlation (the stock and bond prices move in the same direction as the dividend yield and bond yield move in the same direction). In that sense, the model tells us that we *cannot* use the stock-bond correlation to learn about which shocks were more important during that period. We probably can learn something from looking at other moments, but that is not highlighted in the paper.

- In the 2001-2019 calibration, all three shocks lead to a *negative* stock-bond return correlation, again making it difficult to use the stock-bond correlation to learn about shocks.
- The author emphasizes the result that with supply shocks, it is possible to avoid a strongly positive stock-bond return correlation post-covid if monetary policy is of the 2000s style. However, there is a growing literature showing that covid represented both a demand and a supply shock. Based on Figure 6, the stock-bond return correlation would also not be strongly positive with demand shocks under the 2000s style monetary policy. In that sense, the stock-bond return correlation can teach us something about the market's perceived monetary policy rule, not the nature of the shocks expected to be prevalent.

In other work (BauerPfluegerSunderam_rules_Oct2023.pdf (cpflueger.github.io)), the author has a much more direct method based on surveys for assessing the perceived monetary policy in real time. The current approach differs from the survey approach in that (1) the current method does not allow for estimates of the Taylor rule parameters in real-time, but (2) because asset prices are forward-looking, the current method teaches us about the perceived monetary policy rule over the life of the assets.

I would suggest that the author pitches the paper as an approach to learn about the monetary policy rule, rather than an approach to learn about both the MP rule and the nature of the underlying shocks.

(3) Elaborate on stock market reaction to shocks

The paper generally focuses on bonds, nominal and real. Stocks are relevant only because they do well in the good state of the economy so one can assess the beta of bonds with the SDF from the beta of bonds with the stock market (I think this is right, it would be good to clarify this in the paper). However, looking at the middle column of graphs in Figure 6 and as summarized above, what flips the sign of the stock-bond return correlation generated by supply shocks is that the sign of the stock market reaction flips as a function of the monetary policy regime. The stock market impulse response to supply shocks should thus receive more attention that it does. It would be useful to decompose this impulse into components due to news about the yield curve, the equity premium and future dividends. I suspect that all three components react more favorably (from an investor perspective) under the 2000s monetary policy rule.

(4) Calibration/fit

From Table 2, the nominal bond-stock beta in the model is much higher that in the data over the 1979-2001 period. Can this be improved?

(5) Suggestions for the exposition

- The introduction starts by discussing the correlation of stock returns with bond *yields* and then switches to the correlation of stock returns with bond *returns*. It would be easier to follow the argument if the stock-bond return correlation was used throughout.
- Footnote 2 and the note under Figure 1 is a lot of detail for the introduction and repeats the same information. Consolidate and shorten.

- Focus Figure 1 on the stock-bond return correlation for nominal bonds. It is a lot to process both the time series changes and the nominal versus real in the introduction. Then get to the nominal versus real distinction later as part of the analysis. Related, in Figure 6, it would be informative to add a row showing the real yield impulses and relate this back to the stock-TIPS return correlation.
- Clarify what is meant by "prevalent shocks". I think it is the expected mix of shocks going forward but this was not clear.
- Elaborate on what is new in the model relative to Campbell, Pflueger and Viceira (2020). The text states "Different from this prior work, I model supply and demand shocks and focus on how these economic shocks interact with monetary policy to drive bond risks." The model and solution method looks very similar. Is the main model novelty the Taylor rule? Equation (21) in that paper (Macroeconomic Drivers of Bond and Equity Risks (uchicago.edu)) looks very similar to a Taylor rule. Or is the main novelty the application to the post-covid period?